

LAB 09

MULTIPLEXER & DEMULTIPLEXER

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GATE-LEVEL MODELING

- Verilog HDL supports built-in primitive gates modeling.
- The multiple-input gates supported are: **and**, **nand**, **or**, **nor**, **xor**, and **xnor** whose number of inputs are two or more, and has only one output.
- The gate-level modeling is useful when a circuit is a simple combinational.
- For example: **xor** gate using and, or & not gates
- Modeling in Gate level is very difficult for a complex circuit design

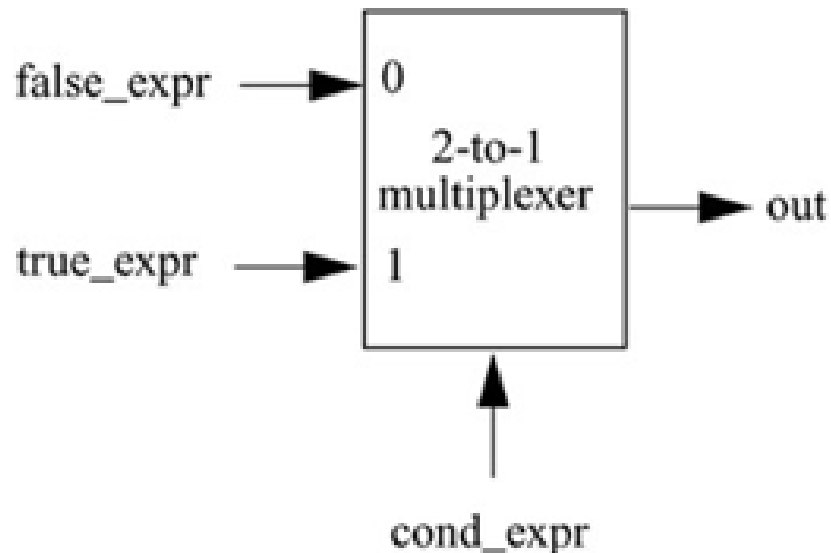
DATA-FLOW MODELING

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- Dataflow modeling style is mainly used to describe complex combinational circuits. Example: 4-bit adder.
- The basic mechanism used is the **continuous assignment**.
- In a continuous assignment, a value is assigned to a data type called net.
- **assign** operator
- `assign out1 = in1 & in2;`
- No clock or trigger is required
- Value of 'out1' immediately changes, if 'in1' or 'in2' value changes.

DATA-FLOW MODELING

- **Conditional operator**
- The conditional operator (`? :`) takes three operands.
- Usage:
- `condition_expr ? true_expr : false_expr;`



```
result = (marks > 33) ? 'p' : 'f';
```

Diagram illustrating the execution of the conditional operator for the code snippet: `result = (marks > 33) ? 'p' : 'f';`

The expression `(marks > 33)` is evaluated. If the result is **FALSE**, the value `'f'` is assigned to `result`. If the result is **TRUE**, the value `'p'` is assigned to `result`.

LAB 09: INTRODUCTION

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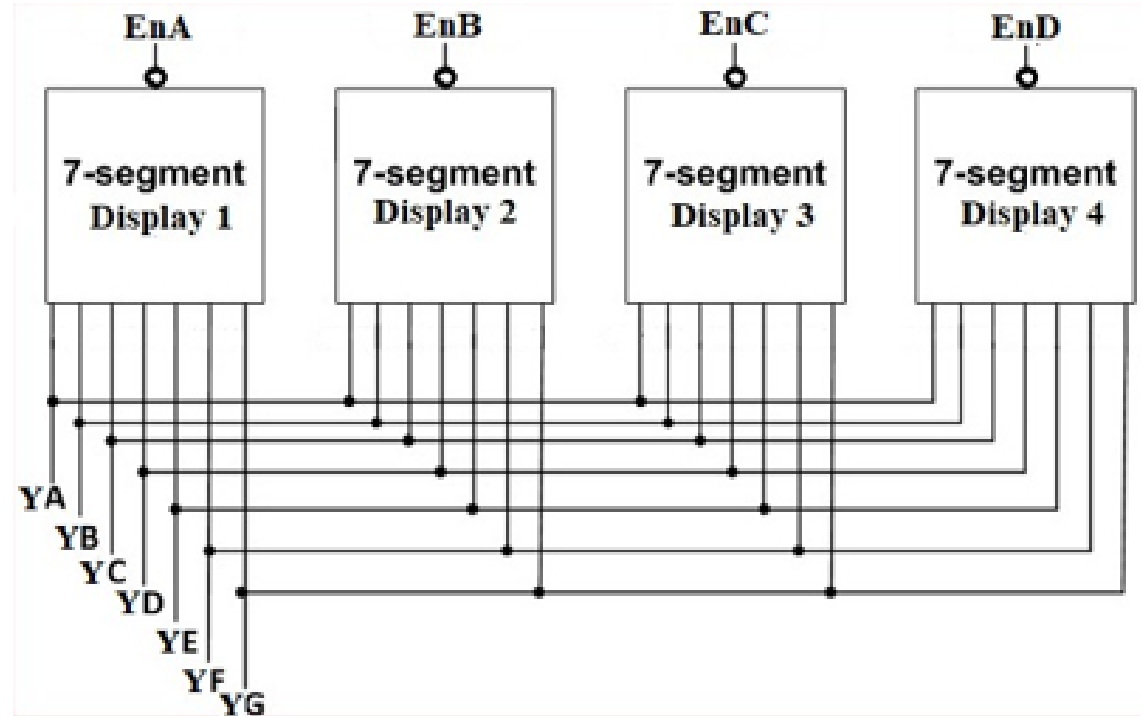


Figure 9.1

The overall system accepts 4-bit data in four data buses A, B, C and D for each seven-segment display, 2-bit data for the select line S to control the 4-bit data, which is passed to the binary to hexadecimal decoder1, the final output (7-bit data) of which goes to the seven-segment display and the seven segment LEDs are controlled by Enable bit to enable or disable the display of selected seven segment display.

LAB 09: INTRODUCTION

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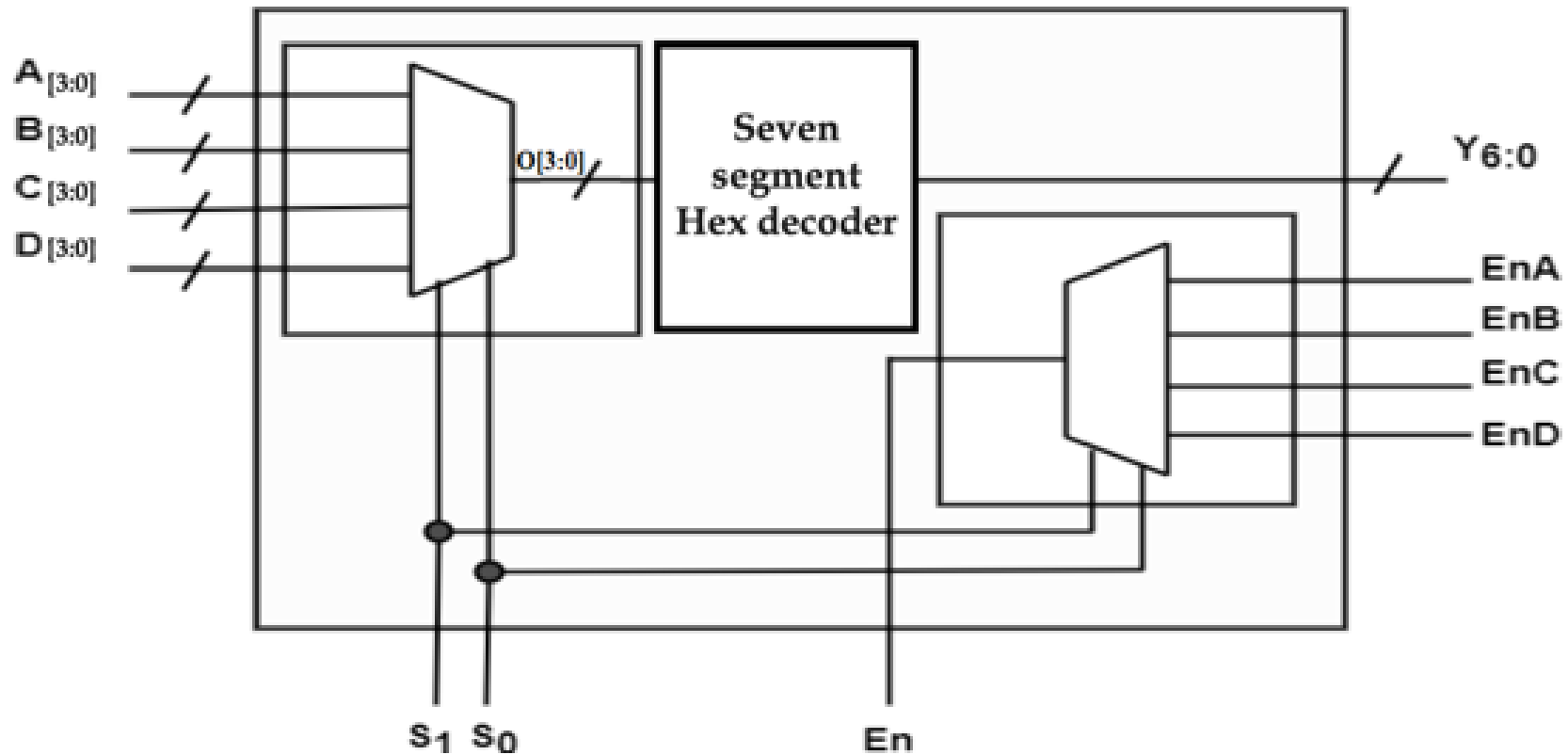
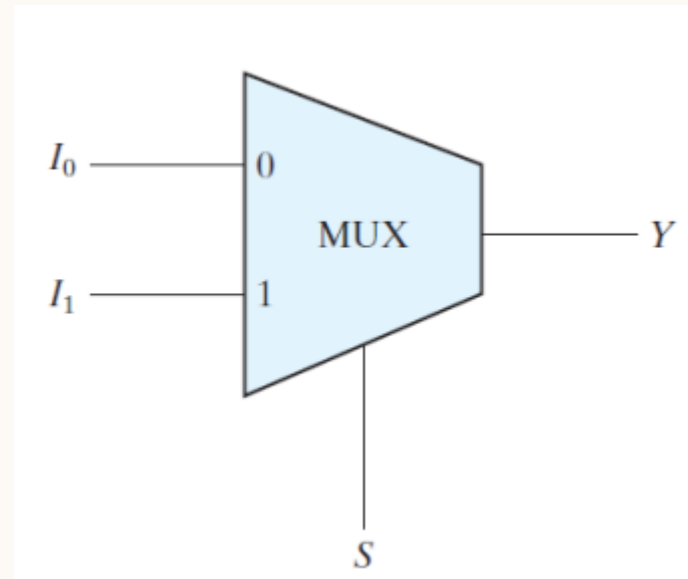


Figure 9.2 Top Level Module

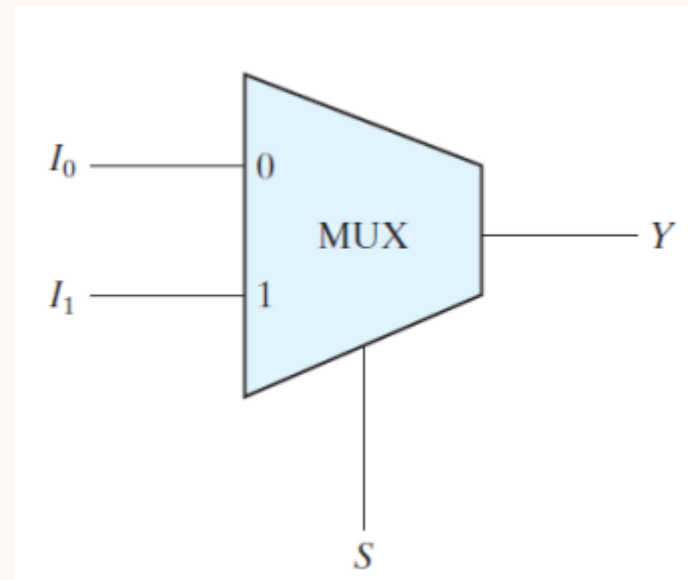
MULTIPLEXER

- In electronics, a multiplexer, also known as a data selector.
- A device that selects between several analog or digital input signals and forwards the selected input to a single output line.



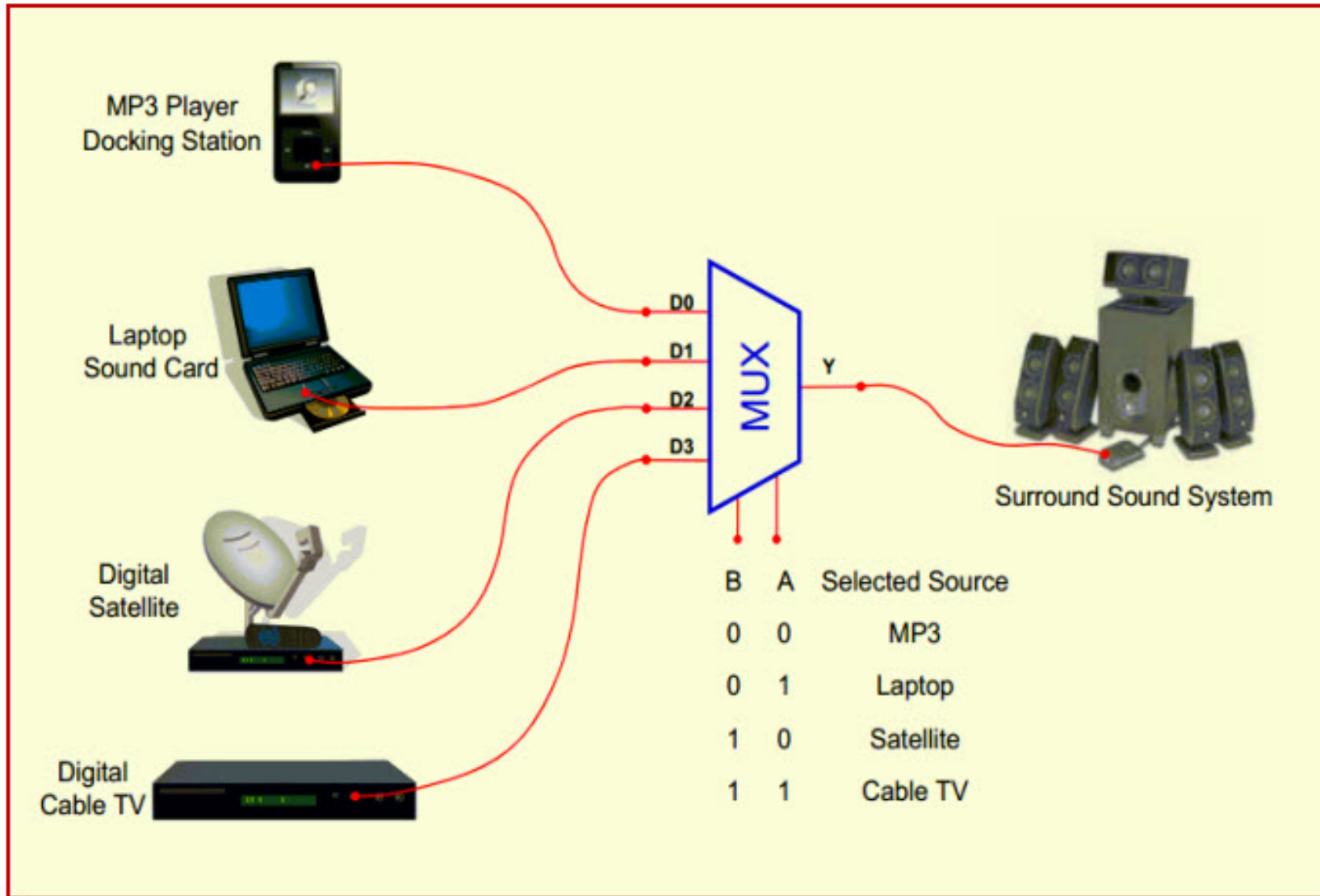
MULTIPLEXER

- **2-to-1 mux**
assign $Y = S ? I_1 : I_0;$

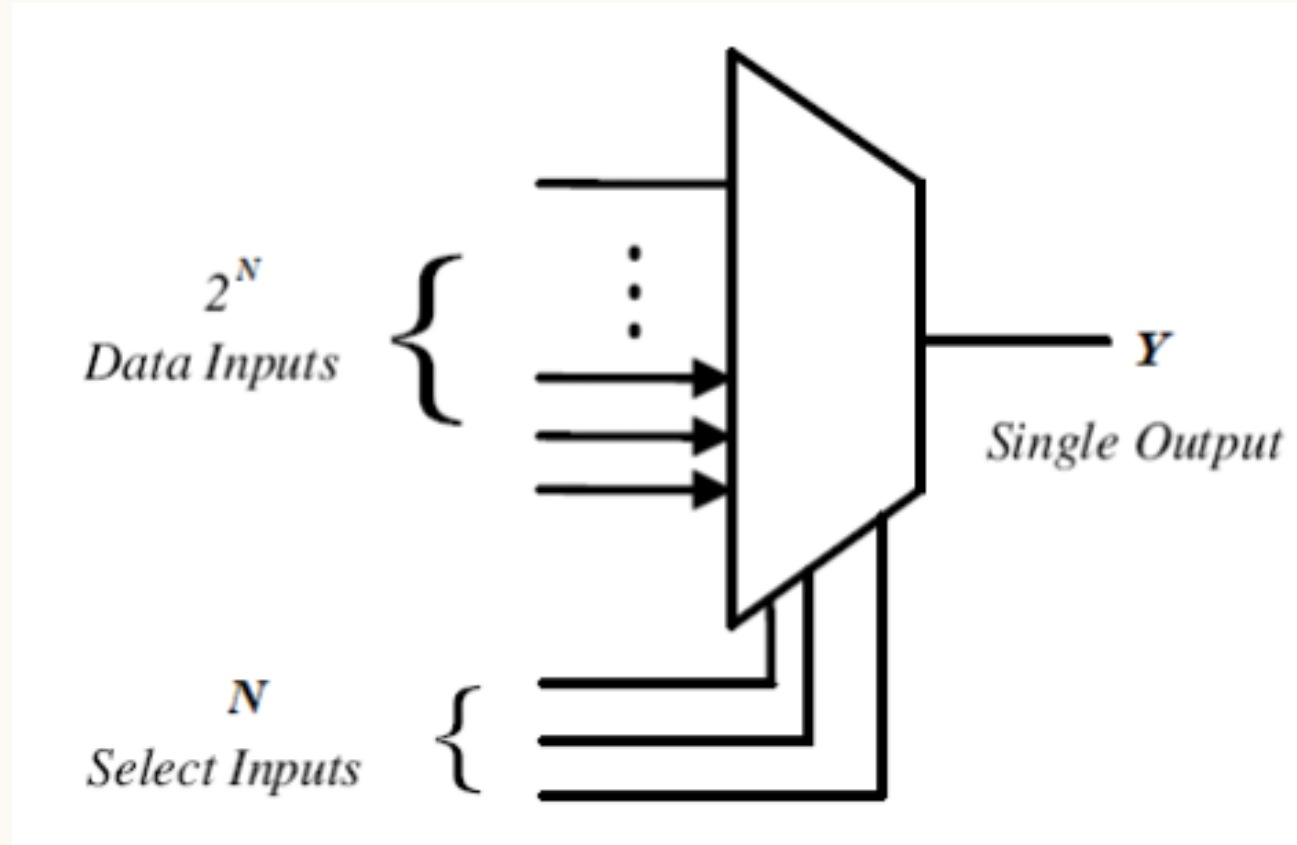


MULTIPLEXER

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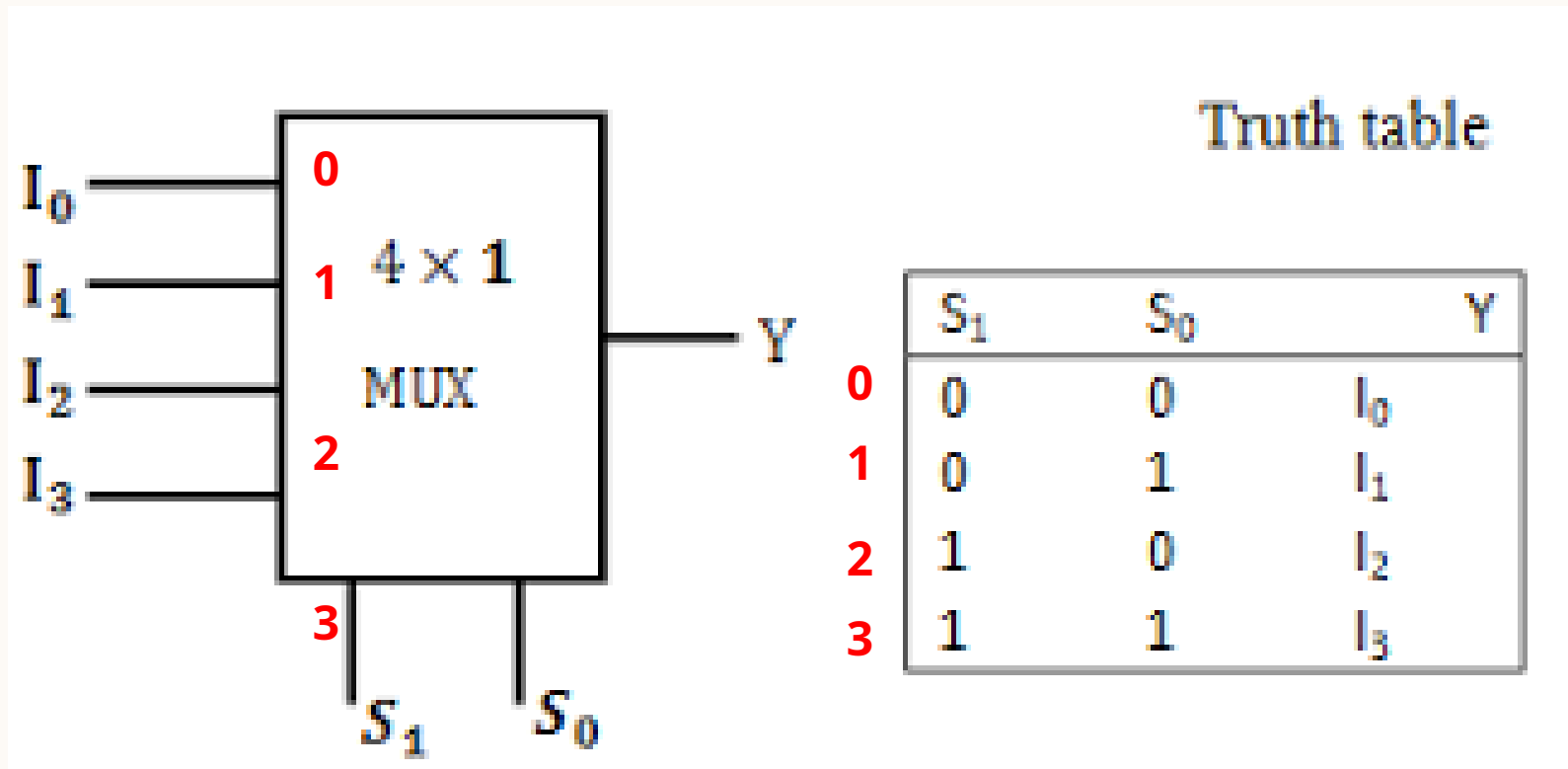


MULTIPLEXER



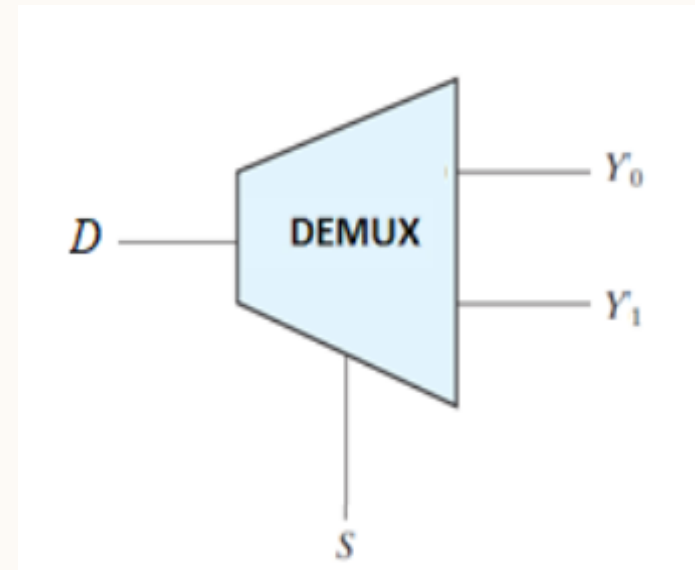
- # of i/p lines = 2^n
- n=select lines

MULTIPLEXER (TRUTH TABLE)

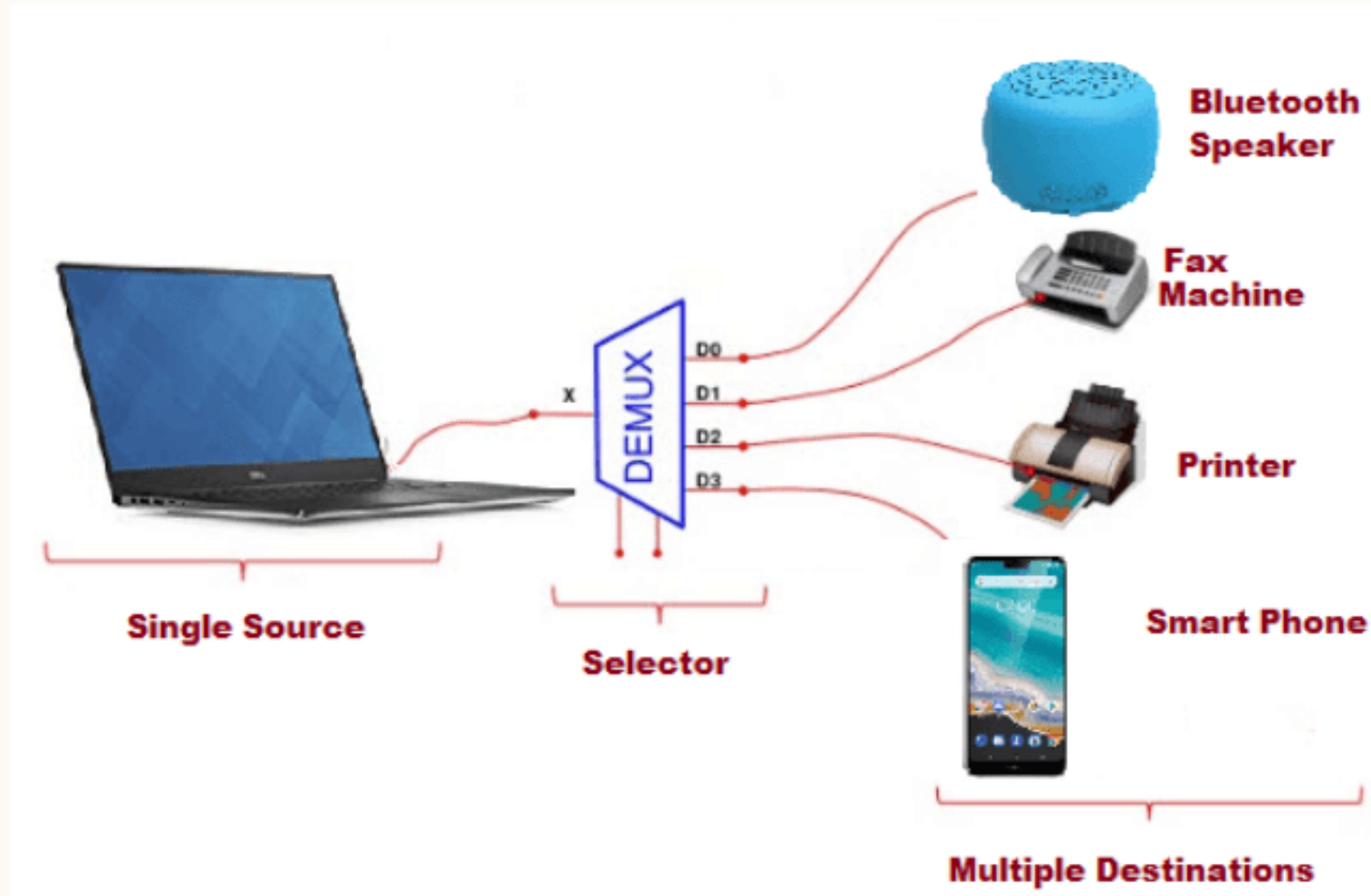


DEMULTIPLEXER

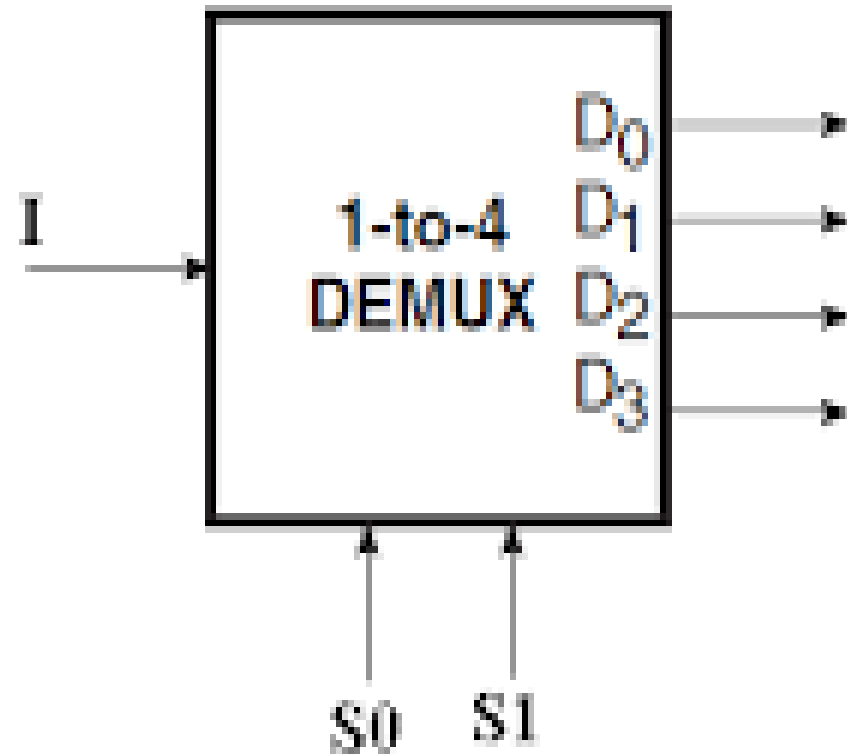
- A Demultiplexer receives the information on a single input line and transmits the same information over one of 'n' possible output lines.
- Single input (Data D)
- Multiple outputs
- Select line – selects the output line to which data will be distributed.



DEMULTIPLEXER

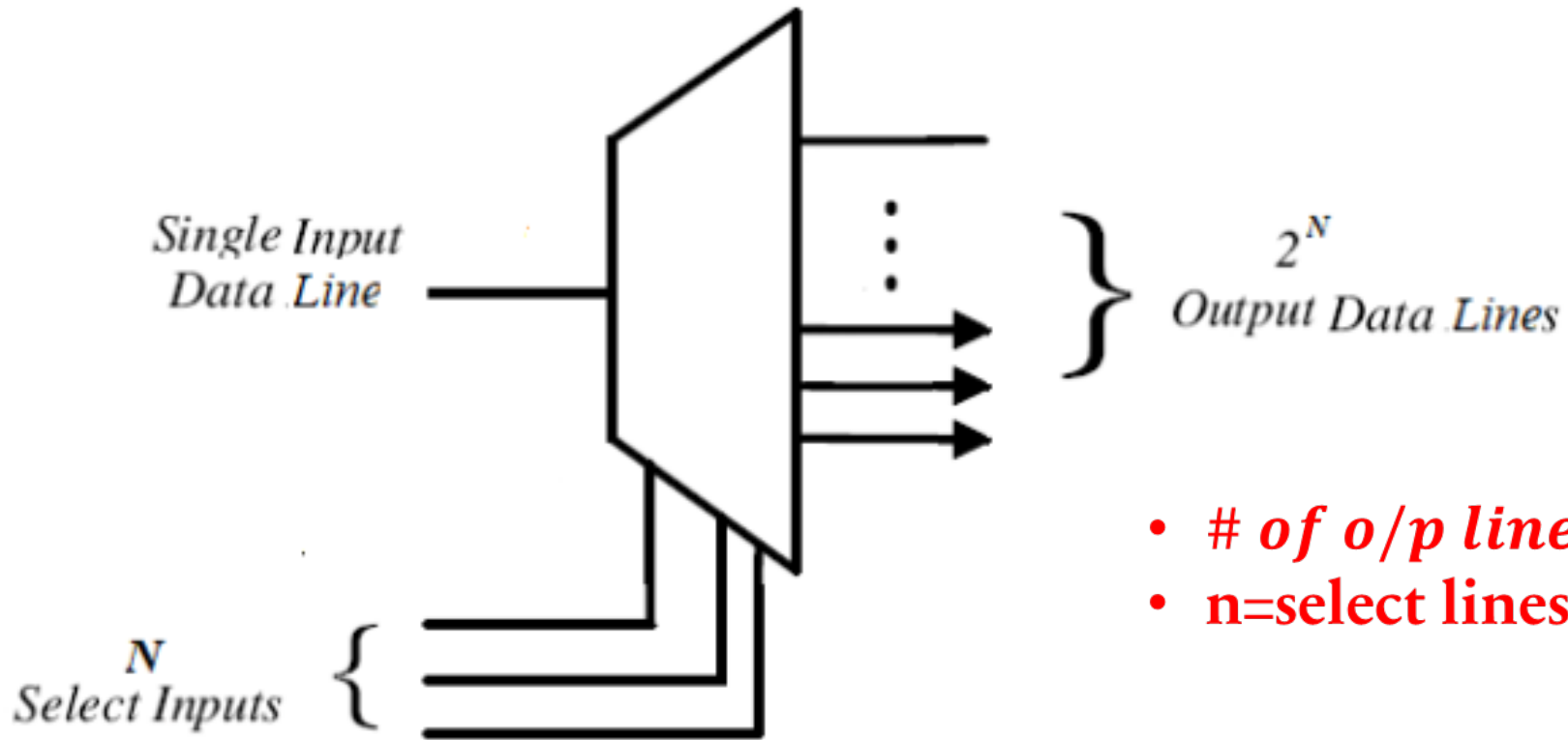


DEMULTIPLEXER (TRUTH TABLE)

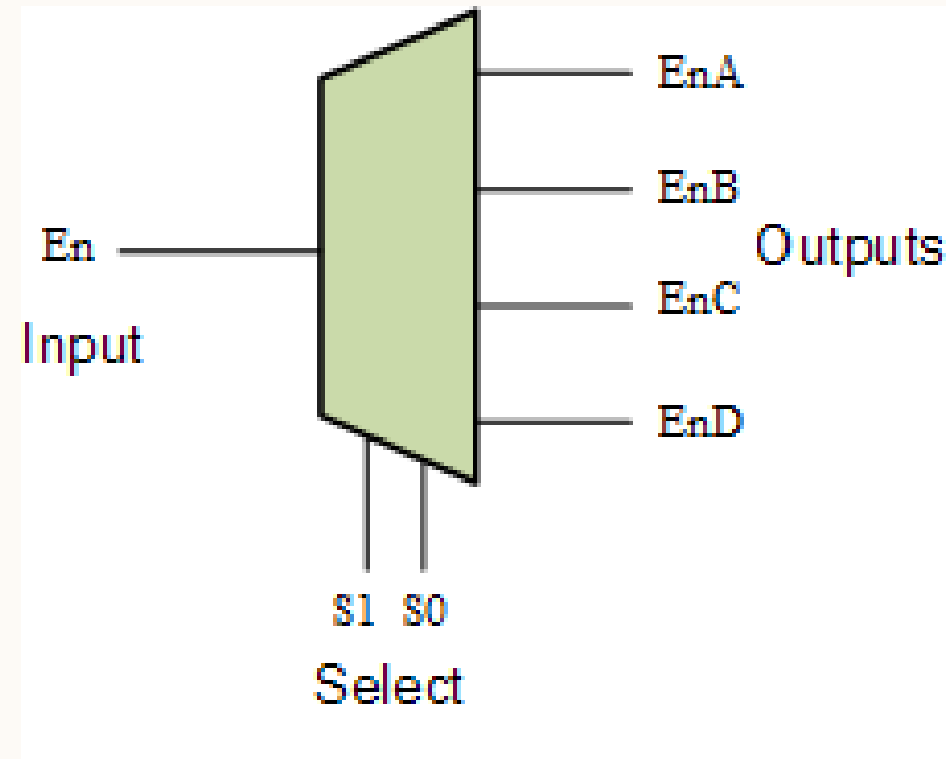


I	Select		O/P			
	S0	S1	D0	D1	D2	D3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

DEMULTIPLEXER



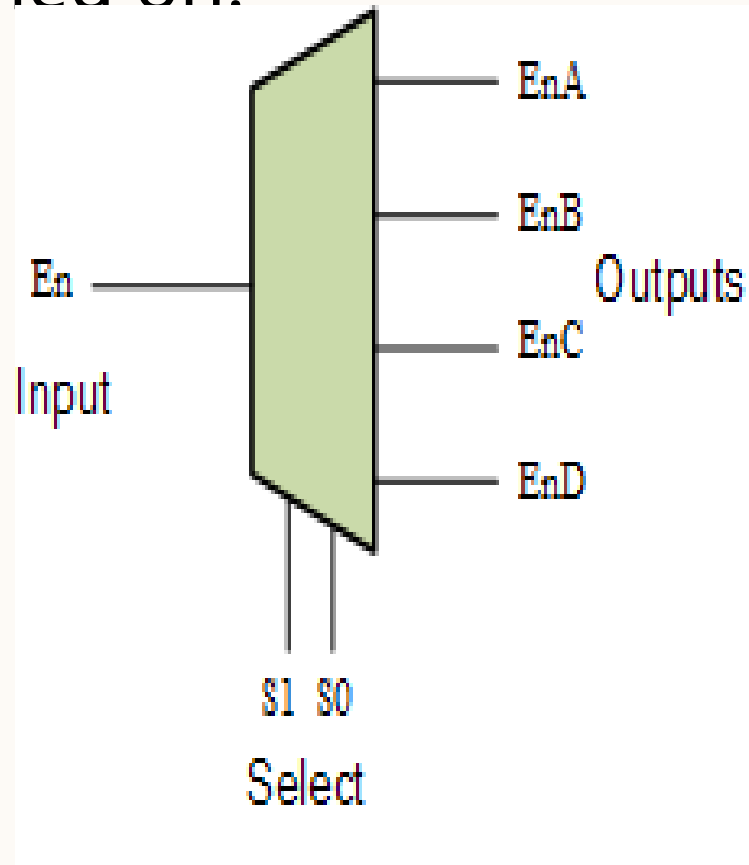
DEMULTIPLEXER



```
assign EnA= (S==2'b00) ? En: 1'b1;
```

TASK - DEMULTIPLEXER

Seven segments are Common Anode i.e, active low, so when you make Enable pin '0' the corresponding seven segment will be turned on.



TOP LEVEL MODULE

